

From tropical forest-carbon measurement to annual carbon accounts

Stock-to-flux identities, census calibration and a historical-data audit

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Abstract

Tropical forest-carbon recalibration changes annual accounts through spatial selection, census increments and release histories. This research article combines a published-data audit with exact stock-to-flux and source–sink identities. The historical 723.97 PgC reassessment implies 472.19 MgC ha^{−1} in its two highest-carbon classes. A nested-mask comparison supplies a conditional lower correction bound; the larger whole-domain density contrast requires an additional spatial assumption. Truncation reproduces the published percentage column and historical endpoints, while both truncation and nearest-rounding models admit a common normalization factor near 0.99016. Crossed class intervals remain an unresolved metadata issue. Carbon correction transfers to emissions through its covariance with release weights. Oppositely ordered corrections and initial combustion fractions suppress a single cohort’s first-year multiplier, but annual mixtures also depend on disturbance history; stationary clearing with complete release recovers the stock multiplier. General paired accounting includes zero-stock regrowth and distinct gain and mortality operators. Exact quadratic and power-law criteria identify when size-dependent recalibration amplifies losses relative to gains, with a geometric-mean criterion in the near-linear limit. EDGAR’s reported 2025 reference is represented by disjoint land categories alongside its separate non-land gas composition. The historical emissions percentage remains conditional on unmatched stock and activity provenance. The analysis provides reproducible diagnostics and measurable conditions for changing attribution, without imposing a universal forest-carbon correction.

Keywords: tropical forest carbon; measurement and accounting; measurement bias; stock-to-flux transfer; secondary-forest regrowth; census carbon balance; size-dependent calibration; source–sink reconciliation; historical evidence audit.

1 Introduction

An emissions diagram combines biological measurement, accounting classification and normalization. These operations are not interchangeable. A larger standing carbon stock need not produce a proportionately larger annual release, and a larger gross release need not imply an equally large increase in net land exchange. Conversely, agreement in a net carbon budget does not establish that each underlying gross component is accurately measured. These distinctions determine what can legitimately be inferred from a revised forest-carbon model.

The historical forest-carbon reassessment [1] motivated three interpretations of the apparent greenhouse-gas source hierarchy: an underestimated total, redistribution within a retained total, or joint revision of sources and removals. The subsequent Amazon essay developed the broader interpretation [2]. The scientific question is not simply whether a larger forest numerator changes a pie chart. It is whether

independently supported biological recalibration changes the physical flows, the accounting allocation, or both.

In the present analysis, the 723.97 PgC estimate is retained as a testable historical hypothesis, not endorsed as a validated pantropical inventory or as sufficient evidence for revising climate models. The measurement question does not require accepting that extrapolation: an argument for reassessing measurement foundations can remain important when its original numerical implementation has unresolved problems. Contemporary evidence from tree reconstruction, destructive reference measurements and locally calibrated allometries makes the measurement question testable [3–6]. Such studies must nevertheless retain their geographic domains, response variables and reference methods; they cannot be pooled informally into a pantropical correction factor.

This article makes three connected contributions. First, it states an exact transfer identity linking spatially varying stock corrections to annual release weights. Second, it separates general paired reconciliation from stock-linked transfers, regrowth increments and size-dependent census corrections, deriving exact directional criteria and a release-timing result. Third, it audits the historical stock, area and percentage inputs rather than treating them as interchangeable quantities. Appendix A gives the standard statistical tools supporting the biological and accounting analysis.

The diagrams in Figs. 1 and 2 distinguish the reported account from a forest-carbon remeasurement framework. Their sector–activity–gas organization adapts the WRI/Ecofys visual convention [7, 8]; intermediate activity nodes are conceptual pathways, rather than EDGAR’s official sector classification. The hatched forest branch denotes an unresolved adjustment, while a separate inset identifies the historical arithmetic and its assumptions.

Scope of inference. The article establishes conditional algebraic results, audits published numbers, and specifies empirical tests. It does not remeasure the historical carbon map, identify a new 2025 forest-emissions percentage, or infer that another sector’s physical emissions fall when its share changes. Stocks, gross transfers, net exchange and gas-equivalent composition are kept distinct throughout. Numerical tests verify the stated calculations; field and inventory validation require independent observations.

2 Scope, evidence status and accounting boundaries

2.1 Research design and evidence status

The evidence selection is purposive, not a systematic review or meta-analysis. It includes the motivating preprint, the original emissions chart, primary measurement studies, forest plot-network analyses and official emissions documentation. Three statuses remain distinct: source-reported quantities, calculations from specified inputs, and proposed empirical tests. Its empirical component is a published-data audit; the analytical component derives and tests conditional identities.

The accounting reference year is 2025 and the source-verification date is 16 September 2026. A publication year does not identify an observation year. EDGAR’s 2026 release reports inventory estimates through 2025 [9]; the final *Global Carbon Budget 2025* article was published in 2026 and retains a projection for the 2025 land-use flux [10]. Source locations and evidence limits are recorded in the public repository’s `data/source_provenance.csv` [11].

2.2 Stocks, transfers and gas-equivalent quantities

Aboveground biomass is dry biological mass; aboveground carbon is the carbon contained in that mass. Although 1 Pg = 1 Gt, a stock in PgC cannot be inserted into a flow in GtC yr^{−1}. For carbon actually emitted as CO₂, the conversion is

$$E_{\text{CO}_2} = (44/12)E_C. \quad (1)$$

It preserves, rather than creates, the time dimension. Movement from live wood to dead wood or harvested products is not necessarily an atmospheric emission in the same year. Carbon pools, transfers and eventual release therefore require explicit representation [12].

A multi-gas inventory instead applies a selected climate-equivalence convention,

$$T_{\text{GHG}} = \sum_g \text{GWP}_{100,g} E_g. \quad (2)$$

This is not a conserved carbon-mass balance. A CO₂ component can legitimately form the numerator of a CO₂e share because the equivalence factor for CO₂ is one; the denominator must still specify the other gases, time horizon and accounting boundary. Gas conversion does not make a net land-use category equivalent to gross tropical deforestation.

Throughout, a *gross component* means an explicitly positive emission or removal within the selected account. It does not mean that an anthropogenic inventory includes all ecosystem respiration or gross primary production. Those ecosystem fluxes cannot be added indiscriminately to an inventory of human-induced emissions.

2.3 The contemporary reference

Table 1: Reported reference quantities through 2025. Values are rounded source estimates or projections, not recalibrated forest-carbon results. LULUCF denotes land use, land-use change and forestry.

Quantity	Annual value		Boundary and status
EDGAR GHG excluding LULUCF	54.1 GtCO ₂ e		2025 inventory; AR5 GWP ₁₀₀
EDGAR GHG including net LULUCF	Approximately 55.0 GtCO ₂ e		Same reported account including its land balance
EDGAR Forest Land	−5.5 GtCO ₂		Living-biomass net removal; remaining forest and land converted to forest
EDGAR Deforestation	+3.7 GtCO ₂		Net category, all regions; includes tropical forest-fire CO ₂
EDGAR Fires	Approximately 2.2 GtCO ₂ e		Allocated fire account; tropical forest-fire CO ₂ excluded to avoid overlap
EDGAR Organic Soils	About +1.1 GtCO ₂		Reported CO ₂ subtotal; the category also includes N ₂ O
EDGAR Other	See signed balance		Non-biomass forest pools and remaining land categories; not independently extracted here
EDGAR net LULUCF	About +0.9 GtCO ₂ e		Sum across the five categories above
GCB net land-use CO ₂	About +4.1 GtCO ₂		2025 projection, low confidence; different attribution convention

The land entries in Table 1 follow the *booklet*, its Section 3, Figure 6 and Annex 2, rather than a press-release paraphrase [9]. Forest Land already includes land converted to forest; afforestation and regrowth must not be appended as a second removal subtotal. Tropical forest-fire CO₂ is assigned to Deforestation, whereas its non-CO₂ gases remain in Fires. Crop-burning CH₄ and N₂O are excluded from Fires where assigned to agriculture. The fire account does not distinguish managed from unmanaged land. These are inventory allocation rules, not a decomposition into all ecosystem gross fluxes.

The rounded entries yield the diagnostic remainder $0.9 - [3.7 - 5.5 + 2.2 + 1.1] \approx -0.6 \text{ GtCO}_2\text{e yr}^{-1}$. Because the organic-soil value quoted in the text is a CO₂ subtotal and the inputs are rounded, this is *not* an independently recovered numerical estimate of Other; it also absorbs omitted subcomponents and rounding. Figure 1 keeps the five categories distinct and labels this remainder accordingly. Land flows do

not feed the fossil-CO₂ block. The non-LULUCF mixture is 73.8% fossil CO₂, 17.6% CH₄, 5.3% N₂O and 3.3% fluorinated gases; the power-and-heat category contributes nearly 30% of the non-LULUCF total [9].

The forest/deforestation estimates are a global, geographically explicit implementation of the IPCC Tier 1 Gain/Loss approach. They combine land-change information with default growth, biomass-expansion, wood-density, carbon-density and root-to-shoot parameters. Annex 2 describes forest-cover inputs through 2022 with subsequent estimates described by the booklet as interpolation, and a deforestation-driver product for 2001–2021; the label “2025 estimate” does not imply that every input was measured in 2025. Other additionally draws on country-reported data. This provenance is relevant to the same density-compatibility scrutiny applied to the historical benchmark [9].

The Global Carbon Budget uses a different attribution of anthropogenic land effects and environmentally induced uptake [10, 13]. Neither its net land-use flux nor the two EDGAR forest entries can be substituted for a resolved pair of gross atmospheric transfers. No numerical paired correction is imposed on those entries without matching pools, processes and attribution conventions.

3 Biological calibration and the plausibility of the historical stock

3.1 What constitutes independent validation?

A dimensionally explicit tree-carbon representation is

$$C_j = \sum_o V_{jo} \rho_{jo}^b f_{jo}^C, \quad (3)$$

where non-overlapping compartments o have green material volume V , basic density ρ^b defined as oven-dry mass per green volume, and dry-mass carbon fraction f^C . With consistent units, the product is carbon mass. Hollow space, bark, leaves and branches must not be double counted; roots do not belong in an aboveground-only target. Moisture conventions must be matched rather than multiplying hydrated mass by a dry-mass carbon fraction.

Modern pantropical allometries explicitly incorporate size, wood density and, where available, height [14]. Structural-bias testing concerns residual or transported errors, not a claim that those covariates are universally ignored. Architecture and tissue chemistry can cause real deviations from an allometric reference. They do not guarantee that every adjustment is upward. The IPCC tropical aboveground default carbon fraction is already 0.47, with scope-specific qualifications, rather than universally 0.50 [12]. Measurements of tropical tree carbon fractions also show that substituting measured fractions for a conventional value can lower estimated carbon [15]. Geometric, density and chemical corrections should consequently be assessed jointly.

The original reassessment combines three-dimensional modelling, tissue measurements and neural-network calibration [1]. A neural network trained on a geometric model’s outputs may emulate that model accurately without independently validating its carbon mass. The same distinction applies when remote-sensing products are calibrated against allometric plot labels. Increasing the number of predicted pixels does not remove an error shared by their training targets. Destructive reference masses, independent material-volume checks and suitable chemistry measurements provide different tests of the chain.

3.2 Area-normalized audit of 723.97 PgC

Table 3 of the preprint reports 723.97 PgC over 2,799,053,606.91 ha. Its classes include high-carbon forest, disturbed vegetation, woodland, scrub, savanna and degraded land; the denominator is therefore a mixed vegetation domain, not a rainforest-only area [1]. For a stock C in PgC, area A in hectares and assumed dry-mass carbon fraction f_C ,

$$\bar{c} = 10^9 C / A, \quad \bar{B} = \bar{c} / f_C. \quad (4)$$

Using the source's displayed totals gives $\bar{c} = 258.65 \text{ MgC ha}^{-1}$. The two highest-carbon classes contain 566.79 PgC over 1,200,345,923.20 ha, implying $472.19 \text{ MgC ha}^{-1}$. Those carbon densities do not depend on any biomass back-conversion.

The original method reports measured tissue-carbon fractions of 0.4111–0.4999 and predicts carbon directly through its geometric and neural-network calibration [1], rather than specifying a uniform pantropical factor of 0.50. Reconstructing the biomass actually implied by that method requires the associated biomass weights: for compartments j , $B_\Sigma = \sum_j C_j/f_j$ and $f_{\text{eff}} = C_\Sigma/B_\Sigma$. The published aggregate does not recover these weights or an effective pantropical fraction. Neither inversion by a tissue-average fraction nor division by 0.50 reproduces the original biomass in their absence.

As explicit sensitivities, $f_C = 0.47$ gives $550.32 \text{ Mg dry biomass ha}^{-1}$ for the mixed domain and 1004.66 for the high-carbon classes; $f_C = 0.50$ gives 517.30 and 944.38 , respectively. If, additionally, all mapped compartments had fractions within the original measured range, their means would lie between 517.40 – 629.16 and 944.57 – 1148.60 , respectively. These conditional conversion bounds are not empirical uncertainty intervals or a recovered biomass surface. The high-class density burden therefore remains substantial, but “1005” is identified as an IPCC-default sensitivity, not the unique implication of the preprint.

Table 2: Published-digit audit of the historical source's Table 3 [1]. Intervals, printed means and percentages are transcribed; the ratio column is $10^9 C/A$. Carbon-density units are MgC ha^{-1} . Areas are shown in million hectares here, with the full two-decimal-hectare values in the repository. Classes 2 and 3 fit each other's intervals. The printed total is 723.97 PgC .

Class	Printed interval	Area (10^6 ha)	PgC	Printed mean	Ratio	Printed %
5	478–605	531.449	285.03	531.05	536.33	39.37
4	336–478	668.897	281.76	417.09	421.23	38.91
2	58–183	419.412	105.54	249.16	251.64	14.57
3	183–336	350.689	40.66	114.80	115.94	5.61
1	1–58	828.607	10.96	13.10	13.23	1.51

The class-2 and class-3 inconsistency has a specific crossed pattern. Class 2's recalculated density, $251.64 \text{ MgC ha}^{-1}$, lies inside class 3's printed interval of 183–336, while class 3's recalculated 115.94 lies inside class 2's interval of 58–183. Their printed means, 249.16 and 114.80 , also satisfy the opposite intervals. A transposition of interval metadata, or of the numerical blocks attached to those intervals, is therefore a concrete explanation to test. Merely changing the order of complete rows would not repair their within-row inconsistency. The published vegetation descriptions also belong to the class metadata and must be checked against the classified raster before a correction is chosen.

The mean-density discrepancy has a common pattern. Define $q_j = \bar{c}_{j,\text{printed}}/(10^9 C_j/A_j)$. The five values in source order are approximately 0.990163, 0.990169, 0.990153, 0.990139 and 0.990397. Under nearest rounding, a value printed to two decimals represents an interval with half-width 0.005; under positive-number truncation, it represents the interval from the printed value to that value plus 0.01. Applying the respective interval rules to stock, area and mean gives common-factor bounds $[0.990140, 0.990190]$ and $[0.990134, 0.990182]$. Both models therefore accommodate one shared normalization. Exact division by 1.01 is excluded by classes 4 and 5 under either rule; the available digits identify an approximately one-percent convention difference, rather than its computational origin.

There is direct evidence for truncation in the percentage column. Truncating $100C_j/723.97$ to two decimals reproduces all five percentages in Table 2; nearest rounding does not reproduce classes 4, 2 and 3. The stock sum is 723.95 PgC against the printed 723.97 , and the area sum is $2,799,053,606.89 \text{ ha}$ against

the printed 2,799,053,606.91. Both discrepancies are two units of the last displayed digit. These results are consistent with truncating component values and a separately computed total.

A probability calculation can compare display conventions only after its assumptions are specified. For five independent uniform hidden component fractions and a total computed from their exact sum under the same display rule, a positive two-unit gap has probability $11/20 = 0.55$ under truncation and $1/120$ under nearest rounding; the probability of a gap of either sign with magnitude two under nearest rounding is $1/60$. These are conditional reference-model probabilities, not posterior probabilities of the historical workflow. Carbon and area outputs may share computations, so their probabilities are not multiplied. Truncation explains the visible pattern parsimoniously, while the original calculation is needed to identify the actual display convention.

The common-factor and crossed-interval patterns favor intact numerical rows with a shared normalization and transposed interval metadata as a hypothesis to test. The published tabulation is recoverable, but the pre-tabulation aggregation workbook, operational class lookup and classified numerical rasters could not be located among the records available for this audit. Their absence from those records does not establish whether they were lost, retained elsewhere or superseded. Consequently, the deposit reproduces the published-digit audit rather than the original map-production pipeline. Pure interval relabelling preserves totals; an aggregation correction need not. The repository documents both display-error models and preserves all source entries.

Spatial support matters, but it does not remove the density challenge. Feldpausch et al.'s 285.1 PgC estimate uses 1667.5 million hectares of specified forest classes, implying $170.97 \text{ MgC ha}^{-1}$ [16]. For any two stock–area totals $C = A\bar{c}$, their ratio decomposes as

$$\frac{C_A}{C_B} = \frac{A_A}{A_B} \frac{\bar{c}_A}{\bar{c}_B}. \quad (5)$$

The complete historical comparison is approximately $2.539 \approx 1.679 \times 1.513$: both the larger area and the different mean density contribute to the numerical ratio. This is an arithmetic factorization, not a causal attribution of the difference. The broader historical domain includes extensive lower-carbon vegetation, so its mixed-domain mean attenuates the high-class density contrast.

Classes 4 and 5 contain 566.79 PgC on 1200.346 million hectares: 1.988 times Feldpausch's entire stock on 0.720 times its area. Their mean density is 2.762 times the whole-domain comparison mean. This highlights the high-density requirement but is not yet a matched-pixel correction factor. Let H denote those classes and $C_{0,H}$ their conventional carbon. If H is contained in Feldpausch's mask, then $0 < C_{0,H} \leq 285.1$ and

$$r_H = \frac{566.79}{C_{0,H}} \geq \frac{566.79}{285.1} \approx 1.99. \quad (6)$$

The value 2.76 is obtained if H had the conventional whole-domain mean density. If its conventional mean is additionally at least that whole-domain mean, the conditional bracket becomes $1.99 \lesssim r_H \lesssim 2.76$. Without this extra density assumption, 2.76 is a comparison scenario, not an upper bound: a lower conventional local density can require a larger factor. Neither nesting nor that density ordering has been established by raster intersection here. The repository supplies explicit counterexamples to an unconditional finite bracket.

A common-mask analysis is needed to estimate $C_{0,H}$ and separate biological calibration from vegetation selection. The comparison with 228.7 PgC also requires matched pools, vegetation support and period [17]; no associated Baccini area has been recovered for this audit. Equation (5) is therefore evaluated for Feldpausch only, while the Baccini historical endpoint remains a total-stock arithmetic reconstruction.

3.3 Interpretation of the density audit

The high-class carbon density, the mixed vegetation support and the unresolved normalization require independent validation before 723.97 PgC can be used as an inventory estimate. The value is a historical high-stock hypothesis. Plot estimates may share the calibration error being investigated, while clearing and harvest sites have limited representativeness; neither is a hard pantropical ceiling. A decisive test would combine independent material-mass measurements in representative strata, coherent tissue fractions, validated class metadata and integration over a common mask. These are empirical requirements of the comparison, rather than assumptions of the transfer identities.

4 Evidence from harvest, laser scanning and forest networks

Reference methods require the same scrutiny as the allometries under test. Gonzalez de Tanago et al. [3] used nine of 29 harvested trees for reconstruction calibration and 20 for independent validation. Their Equation 3 defines bias by the mean residual divided by mean reference AGB, equivalently

$$b = \frac{\sum_j (\hat{B}_j - B_j^{\text{ref}})}{\sum_j B_j^{\text{ref}}}, \quad \frac{\sum_j B_j^{\text{ref}}}{\sum_j \hat{B}_j} = \frac{1}{1+b}. \quad (7)$$

Thus $1/(1-d)$ is valid for a reported negative bias $b = -d$, but as a ratio of *sample totals*, not the mean of individual replacement factors. Their Table 4 explicitly identifies 20 validation trees and gives endpoint factors of approximately 1.180–1.554. Sampling selected the largest tree in each of 29 plots (mean diameter 73.5 cm across the sampled trees). The reported increase of allometric underestimation with size cautions against extrapolating this large-tree sample to a stand: a smaller-tree-weighted population would have a lower aggregate factor only if the same size-dependent calibration and relevant weighting also apply. The reference AGB itself was calculated from harvested-tree geometric measurements, assigned wood density and small-branch expansion, not by weighing all reference biomass; the authors explicitly acknowledge those limitations. This is informative structural validation, but not independent validation of every density or chemical input. Momo Takoudjou et al. [4] likewise demonstrate the importance of checking reconstruction quality rather than declaring laser-derived models error-free.

The moist-forest local-calibration and miombo findings test different mechanisms and spatial scales [5, 6]. The mixed-domain density ratio of approximately 1.51 lies within the Gonzalez de Tanago aggregate range and the miombo comparison range; numerical overlap across different populations and response variables is not independent validation. For the high-carbon subset, the conditional nesting bound of approximately 1.99 exceeds the moist-forest replacement factors in Table 3, while falling within the miombo range of 1.5–2.2. The approximately 2.76 uniform-density scenario exceeds all listed factors but depends on assigning the comparison's whole-domain mean to the subset. These comparisons locate the validation burden without treating regional effect sizes as interchangeable global multipliers. Conversely, height integration and map fusion can reduce estimated stocks [16, 18]. The evidence supports conditional, heterogeneous corrections, not a predetermined sign at every scale.

At the declared 0.47 conversion sensitivity, the source's mixed-domain mean is roughly 550 Mg ha⁻¹ and its high-class mean is roughly 1005 Mg ha⁻¹. These should be read against both stand-scale and regional evidence. The reported clearing-site range of 292–436 Mg ha⁻¹ includes fuel pools not identical to live AGB [19]; the Amazonian and African network estimates have their own forest selection and allometric dependence [20, 21]. None is a hard biological ceiling. Taken together, however, they do not provide empirical support for the high-class carbon density or its approximately 945–1149 Mg ha⁻¹ conditional source-fraction conversion across that extensive domain.

A genuine whole-hectare destructive mass census would strengthen this comparison, but a hectare-scale clearing experiment must not be relabelled as such solely because its footprint is one hectare.

Table 3: Selected primary evidence and the limit of its transfer to a global correction. Percentages have different denominators and must not be pooled as interchangeable effect sizes. Dry-mass densities are in Mg ha^{-1} .

Study / support	Reported result	Interpretation limit
Gonzalez de Tanago et al. (2018); 29 trees, of which 20 formed the independent validation set	Harvest-geometry reference: TLS bias -3.68% ; pantropical-model biases -15.22 to -35.67% .	Geometry and assigned density, not complete mass weighing. Aggregate validation-sample replacement factors approximately 1.180–1.554.
Momo Takoudjou et al. (2018); 61 harvested trees, Cameroon	Edited quantitative structure models had 4.68% volume bias; TLS- and harvest-calibrated allometric coefficients were statistically indistinguishable.	The quoted percentage is <i>volume</i> , not global carbon bias. Manual model editing and density remain consequential.
Huertas et al. (2025); 118.75 ha of French Guianan plots	Local species-specific ALS height–diameter models produced 11–13% higher AGB estimates.	Local model comparison; not independent weighing of a pantropical carbon stock.
Demol et al. (2024); miombo landscape	Carbon estimates 1.5–2.2 times conventional comparison estimates at landscape scale.	Dry woodland and scaling methods differ from moist-forest and whole-tropics inference.
Kauffman et al. (1995); four slashed Amazonian primary forests	Total aboveground fuel biomass 292–436; forest-floor material is included.	Clearing-site inventories are not equivalent to complete, independent weighing of all live trees in one hectare. Pools differ from live AGB.
Malhi et al. (2006); Amazonian plot-network synthesis	Regional aboveground live biomass approximately 200–350, with spatial variation and additional small-stem/liana accounting.	Allometry-dependent spatial estimates, useful as contextual comparison, not an independent absolute ceiling.
Lewis et al. (2013); 260 intact African plots	Mean AGB 395.7; 95% confidence interval on the mean ± 14.3 .	Intact closed-canopy sample, not all vegetation; reference calibration can be shared.
Feldpausch et al. (2012); height-integrated tropical analysis	Including height reduced the estimated total from 320.5 to 285.1 PgC over the same forest mask.	A counter-direction model result; compared masks must remain identical.
Avitabile et al. (2016); fused pantropical map	Fusion and bias adjustment gave a lower total than the input-map estimates.	Independent map differences can involve support, calibration and fusion, not one universal biological multiplier.

Inventory-derived initial masses and sampled burn consumption remain model-assisted estimates. Here harvested-tree geometry comparisons, any directly weighed reference material, clearing inventories and plot networks remain distinct evidential classes. This distinction prevents an apparently independent benchmark from covertly reusing the calibration being tested.

5 The stock-to-flux transfer identity

5.1 Matched support and fixed release operators

Consider spatial or cohort units $i = 1, \dots, n$ with positive baseline carbon C_i , local correction factors $r_i \geq 0$, and non-negative annual release weights w_i . The units, carbon pools and disturbance histories are held fixed. The weight has dimensions yr^{-1} when a stock is converted to an annual flow. Define

$$\mu_i = \frac{C_i}{\sum_j C_j}, \quad r_C = \sum_i \mu_i r_i, \quad r_F = \frac{\sum_i C_i r_i w_i}{\sum_i C_i w_i}, \quad (8)$$

with $\mathbb{E}_\mu[w] > 0$. Carbon-to- CO_2 conversion cancels in this ratio if applied consistently.

Proposition 5.1 (Stock-to-flux transfer). *Under these conditions,*

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w)}{\mathbb{E}_\mu[w]}. \quad (9)$$

Consequently $r_F = r_C$ if and only if the stock-weighted covariance is zero. Moreover,

$$\min_i r_i \leq r_F \leq \max_i r_i, \quad |r_F - r_C| \leq \sigma_\mu(r) \text{CV}_\mu(w). \quad (10)$$

Proof. Equation (8) gives $r_F = \mathbb{E}_\mu[rw]/\mathbb{E}_\mu[w]$. Substitution of $\mathbb{E}_\mu[rw] = \mathbb{E}_\mu[r]\mathbb{E}_\mu[w] + \text{Cov}_\mu(r, w)$ proves the identity. A positive denominator gives the equivalence. The first bound follows because r_F is a convex combination of r_i with weights proportional to $C_i w_i$. The second follows from Cauchy–Schwarz and $\text{CV}_\mu(w) = \sigma_\mu(w)/\mathbb{E}_\mu[w]$. $\square$

The result is the covariance decomposition of a weighted mean, algebraically identical to the selection term of the Price equation [22]. Its scientific role is to identify the missing empirical quantity: the association between where carbon is misestimated and where that carbon is released. Statistical independence is sufficient but not necessary; zero covariance under the specified stock weights is the exact condition. A constant r_i is another sufficient condition. Neither condition follows from a ratio of published global totals on different masks.

Table 4: Two-unit counterexample. Baseline stocks are (50, 50) and corrections (1, 3) in both rows, so $r_C = 2$. Only the annual release location changes. Units are synthetic.

Annual release weights	$\mathbb{E}_\mu[w]$	$\text{Cov}_\mu(r, w)$	r_C	r_F
(0.02, 0)	0.01	−0.01	2	1
(0, 0.02)	0.01	+0.01	2	3

5.2 Census changes, offsets and state-dependent calibration

For a stable scalar correction, two-census stock change obeys $\Delta C^* = r\Delta C$. A time-varying calibration instead gives the exact decomposition

$$C_1^* - C_0^* = r_0(C_1 - C_0) + C_1(r_1 - r_0). \quad (11)$$

The second term must be interpreted rather than automatically labelled instrument drift. It can reflect a changed calibration procedure, but it can also arise from a fixed size- or age-dependent calibration evaluated at different biological states. A fixed additive offset cancels for an unchanged set of units; recruitment, mortality or support changes can prevent that cancellation. A stable multiplicative error generally changes stock differences. A stand-wide scalar therefore requires justification for heterogeneous trees and changing composition.

This matters for repeated-census forest sinks. Plot-network analyses estimate biomass dynamics and net live-biomass accumulation using repeated tree observations and biomass relationships [23]. A shared, stable multiplicative mass correction changes these inferred gains and losses, not only disturbance emissions. However, a net increase in live aboveground biomass is not gross primary production or automatically net ecosystem–atmosphere uptake. Mortality transfers mass into other pools; decomposition, roots, soils, respiration and exports are needed to close the atmospheric boundary [12]. The paired formulation below applies only after those quantities have been aligned.

5.3 Disturbance cohorts and carbon fate

For area $A_{ia\tau}$ disturbed by process a in year τ , let h index disjoint tree-size, organ and fate compartments, with pre-disturbance carbon density $c_{iah\tau}$ and affected fraction $d_{iah\tau}$. Set $L_{iah\tau} = A_{ia\tau}c_{iah\tau}d_{iah\tau}$. With $k_{iah}^{\text{CO}_2}(t - \tau)$ the annual release rate (yr^{-1}) for CO_2 in year t ,

$$F_{\text{CO}_2}(t) = \frac{44}{12} 10^{-9} \sum_{i,a,h,\tau \leq t} L_{iah\tau} k_{iah}^{\text{CO}_2}(t - \tau). \quad (12)$$

Here L is in MgC and the output is $\text{GtCO}_2 \text{ yr}^{-1}$. For annual bins, $k\Delta t$ is the released fraction with $\Delta t = 1$ yr; cumulative fate fractions must not exceed the available cohort carbon. Other fate fractions represent retained live biomass, dead material, products, charcoal, exports or carbon released in other gases. Their mass bookkeeping must conserve carbon over the specified horizon. Regrowth is a separate uptake term and cannot be represented by making a gross release kernel negative.

The fixed-operator identity is exact when the correction changes carbon amounts but not areas, disturbance fractions or fate kernels. When those inputs also change, r_F must be recomputed from the revised products rather than attributed exclusively to r_C . An index can include cohort and pool, so the identity is compatible with delayed release; it does not remove the need to measure those delays.

5.4 Size-dependent release and the timing of a stock correction

The opposite-order sign used below is Chebyshev’s sum inequality for similarly or oppositely ordered sequences, here written under stock weights. Its application concerns physical release from a defined cohort, rather than a rule for booking inventory stock losses. Combustion measurements in a rainforest clearing distinguish markedly different completeness among fuel-size components [24]. This supplies a physical candidate for the covariance in Proposition 5.1, not a universal monotone law: diameter, moisture, decay, burn history and exposure must be controlled. Where structural underestimation is concentrated in large wood and those compartments burn less completely, their carbon correction receives less weight in initial emissions. Extraction of boles into retained products can reinforce the initial delay, provided products and exports are tracked rather than lost from the boundary [12].

Proposition 5.2 (Opposite ordering and cumulative release). *On a fixed affected-carbon cohort with weights $\mu_i = C_i/C_\Sigma$, let $p_{it} \geq 0$ be the fraction of compartment i emitted as CO_2 in annual bin t , and $P_i(H) = \sum_{t \leq H} p_{it}$. If $(r_i - r_j)(p_{i0} - p_{j0}) \leq 0$ for every pair, then, for positive initial release, $r_{F,0} \leq r_C$. For a horizon with positive cumulative release,*

$$r_{F,\leq H} = r_C + \frac{\text{Cov}_\mu(r, P(H))}{\mathbb{E}_\mu[P(H)]}. \quad (13)$$

If $P_i(\infty) = 1$ for every compartment, $r_{F,\leq H} \rightarrow r_C$. The same conclusion holds for any common positive limiting fraction.

Proof. The covariance identity

$$\text{Cov}_\mu(r, p_0) = \frac{1}{2} \sum_{i,j} \mu_i \mu_j (r_i - r_j)(p_{i0} - p_{j0}) \quad (14)$$

has non-positive summands under the ordering assumption. Apply [Proposition 5.1](#) to p_0 and to $P(H)$. A constant positive limiting release fraction has zero covariance. Strict initial inequality holds if some positive-weight pair has a strictly negative product. $\square$

The stock reference here is *carbon affected in the matched cohort*, not all standing pantropical carbon. The premise is a joint ordering to be tested, not a consequence of tree size alone; diameter-dependent drying or repeated burning can alter it. Complete oxidation as CO_2 within the tracked boundary is stronger than clearing a stand. Persistent charcoal, retained products, export outside that boundary and other gaseous carbon can leave unequal limiting fractions and nonzero cumulative covariance. A delayed correction need not reappear in every subsequent year. Under complete accounting, its cumulative allocation does recover the stock multiplier.

A synthetic two-compartment example uses stocks (50, 50), corrections (1, 3) and first-year fractions (0.8, 0.1). Then $r_C = 2$ but $r_{F,0} = 55/45 = 1.222\dots$. If the remaining fractions (0.2, 0.9) are subsequently oxidized, their combined later multiplier is $145/55 = 2.636\dots$, and the lifetime multiplier is $200/100 = 2$. These synthetic transfers describe a physical release history. Applying them to EDGAR requires identifying whether the target is conversion-year stock-loss booking or atmospheric release: the inclusion of tropical fire CO_2 in Deforestation does not itself make a decay kernel part of the inventory operator. [Section 5.5](#) distinguishes those accounting conventions.

5.5 Annual mixtures, disturbance trends and inventory conventions

A year- t release account combines cohorts of different ages. Assume fixed compartment shares $C_i > 0$ per unit disturbance, fixed corrections r_i , annual fate fractions $p_{i,s} \geq 0$, and a non-negative disturbance history $D(t-s)$. For annual time bins, let

$$W_i(t) = \sum_{s \geq 0} D(t-s) p_{i,s}, \quad r_{F,t} = \frac{\sum_i C_i r_i W_i(t)}{\sum_i C_i W_i(t)} = r_C + \frac{\text{Cov}_\mu(r, W(t))}{\mathbb{E}_\mu[W(t)]}. \quad (15)$$

Here $C_i D$ is affected carbon, and division by the one-year bin width gives an annual rate. The common stock weights presume stable compartment composition; changing forest types or disturbance selection require a cohort-specific extension.

Corollary 5.3 (Stationary mixtures and a two-lag trend criterion). *If D is constant over all contributing ages and $\sum_s p_{i,s} = 1$ for every compartment, then $r_{F,t} = r_C$. With two lags, $p_{i,1} = 1 - p_{i,0}$, and $D_0 = D(t)$, $D_1 = D(t-1)$,*

$$r_{F,t} - r_C = \frac{(D_0 - D_1) \text{Cov}_\mu(r, p_0)}{D_1 + (D_0 - D_1) \mathbb{E}_\mu[p_0]}. \quad (16)$$

The denominator is assumed positive. Oppositely ordered corrections and initial releases therefore suppress the annual multiplier during increasing disturbance, enhance it during decreasing disturbance, and cancel under stationarity.

Proof. Stationarity and complete ultimate release make each W_i equal to the same constant. For two lags, $W_i = D_1 + (D_0 - D_1)p_{i,0}$. Substitution into (15) proves the formula and sign. $\square$

For the synthetic cohort with $C = (50, 50)$, $r = (1, 3)$ and initial fractions $(0.8, 0.1)$, constant clearing gives annual baseline and corrected releases of 100 and 200. Doubling the latest cohort gives 145 and 255, hence $51/29 \approx 1.759$; halving it gives 77.5 and 172.5, hence $69/31 \approx 2.226$. The single-cohort first-year suppression is therefore a timing effect whose annual expression depends on history. Longer kernels and changing composition require the complete convolution rather than the two-lag sign rule.

An inventory stock-loss convention and an atmospheric-release history are different operators. EDGAR describes a Tier 1 Gain/Loss implementation, but its Annex 2 does not supply a size-resolved decay kernel for the deforestation category [9]. Under a conversion-year convention that assigns biomass losses immediately, the relevant weights are affected fractions and converted area; decomposition timing does not modify that booked category. The release convolution instead matters when comparing the account with actual atmospheric transfers or with (36). A Tier label alone does not recover all implementation details.

Edge-associated biomass contrasts provide an empirical reason to measure the covariance rather than assume it vanishes. Chaplin-Kramer et al. report approximately 25% lower biomass within 500 m of tropical forest edges and estimate nearly 10% overstatement by Tier 1 carbon-stock methods that omit these effects [25]. These findings establish a spatial carbon contrast. If a uniform reference overstates edge stocks and independently observed clearing weights concentrate at those edges, the correction–disturbance covariance is negative. The biomass study alone does not measure that joint clearing distribution or establish a universal global sign.

6 Paired source–sink corrections

6.1 General reconciliation first; common-stock transfer as a special case

Let $F, U > 0$ be a specified source and positive removal magnitude, with corrected quantities supplied by appropriate, potentially different measurement operators. Define $r_F = F^*/F$, $r_U = U^*/U$ and the signed account $N = F - U + Z$.

Proposition 6.1 (Operator-independent net correction). *For any such corrected components,*

$$\boxed{\Delta N = (r_F - 1)F - (r_U - 1)U + \Delta Z.} \quad (17)$$

No common standing-stock basis or size-independent calibration is required.

Proof. Subtract $F - U + Z$ from $F^* - U^* + Z^*$ and substitute the two component ratios. $\square$

A more restrictive representation sets

$$F = \sum_i C_i w_i^+, \quad U = \sum_i C_i w_i^-, \quad w_i^\pm \geq 0, \quad (18)$$

with positive stocks, fixed weights and the same stock correction r_i multiplying both terms *within each unit*. Applying Proposition 5.1 separately gives

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w^+)}{\mathbb{E}_\mu[w^+]}, \quad r_U = r_C + \frac{\text{Cov}_\mu(r, w^-)}{\mathbb{E}_\mu[w^-]}. \quad (19)$$

Consequently, only in that shared-stock fixed-operator case,

$$\Delta N = (r_C - 1)(F - U) + C_\Sigma \text{Cov}_\mu(r, w^+ - w^-) + \Delta Z. \quad (20)$$

The proof is substitution in (17), using $F/\mathbb{E}_\mu[w^+] = U/\mathbb{E}_\mu[w^-] = C_\Sigma$. Spatially heterogeneous r_i remain permissible, but size-dependent change *within* a growing unit cannot generally be replaced by that unit's stock-level multiplier. Positive regrowth from zero stock is outside this representation, and intact-plot increments require endpoint calibration when r changes with state. Thus (17), not (20), governs the general application. Carbon units, atmospheric boundaries and overlap must be reconciled before either is used; a common CO₂ conversion can then be applied.

6.2 Regrowth from zero stock requires an increment basis

Secondary-forest recovery is a trajectory through age, composition and environment, not a fixed fraction of pre-clearing carbon. Neotropical observations document age-dependent and site-dependent biomass recovery after forest removal [26]. Accordingly, let cohort a occupy fixed area A_a , with baseline live-carbon densities c_{a0}, c_{a1} at two ages separated by $\Delta t > 0$. Let $c_{ak}^* = r_{ak}c_{ak}$ denote a specified endpoint recalibration. Areas in hectares and densities in MgC ha^{-1} give gains in MgC yr^{-1} when the interval is in years. For a selected non-decreasing baseline trajectory, define $g_a = (c_{a1} - c_{a0})/\Delta t \geq 0$. Its corrected increment satisfies

$$g_a^* = \frac{r_{a1}c_{a1} - r_{a0}c_{a0}}{\Delta t} = r_{a0}g_a + \frac{c_{a1}(r_{a1} - r_{a0})}{\Delta t}. \quad (21)$$

This is an age-cohort application of the endpoint identity, not evidence of temporal calibration drift. In particular, $c_{a0} = 0 < c_{a1}$ gives $g_a^* = r_{a1}g_a$: the cohort is retained and its initially undefined stock ratio need not be estimated.

Proposition 6.2 (Increment-weighted regrowth correction). *For the fixed cohort areas, let $I_+ = \{a : g_a > 0\}$, $I_0 = \{a : g_a = 0\}$, and $K = \sum_a A_a g_a > 0$. Define $q_a = g_a^*/g_a$ on I_+ and $v_a = A_a g_a/K$. Then*

$$K^* = \sum_{a \in I_+} A_a q_a g_a + K_0^*, \quad \frac{K^*}{K} = \mathbb{E}_v[q] + \frac{K_0^*}{K}, \quad K_0^* = \sum_{a \in I_0} A_a g_a^*. \quad (22)$$

Proof. Partition the sum $K^* = \sum_a A_a g_a^*$ into I_+ and I_0 , and substitute the definition of q_a on the former. The weights v_a sum to one because zero increments contribute nothing to K . $\square$

The averaging weights are *baseline gains*, not initial stocks. If $K_0^* = 0$, the replacement factor is a gain-weighted mean; a uniform endpoint correction supplies the familiar common-scale special case. A cohort with zero baseline increment but nonzero revised increment contributes additively and cannot be discarded. If $K = 0$, a total multiplier is undefined even when K^* is positive. A newly represented pool with zero baseline carbon at both endpoints also needs an additive model rather than a finite multiplicative correction. Baseline losses and corrected sign reversals require explicitly signed accounting instead of calling every term a removal.

For example, a zero-stock hectare that gains 5 MgC over one year gains 6 after a final-stock correction of 1.2; it remains part of the analysis despite having zero initial carbon. These are illustrative numbers. K is live-carbon accumulation over the selected interval, not gross photosynthesis or a complete atmospheric removal. Regrowth cohorts must be disjoint from any intact-forest accumulation subtotal, and mortality transfers, dead pools, soil exchange and exported carbon must be reconciled before K contributes to an atmospheric account.

6.3 Common-error cancellation and its limits

For a uniform stable multiplier r , fixed Z gives

$$\Delta N = (r - 1)(F - U), \quad \eta = \frac{|F - U|}{F + U}, \quad \frac{|\Delta N|}{|\Delta F| + |\Delta U|} = \eta \quad (r \neq 1). \quad (23)$$

η lies between zero and one. Thus a nearly balanced pair can undergo a large revision of both components with a small *absolute* adjustment of their net difference. The relative change of the nonzero calibrated net difference is still $r - 1$; cancellation must not be described as generally small relative net bias. Exact balance leaves the net unchanged, regardless of the common multiplier.

The converse sensitivity is just as important. Write $r_F = r + \delta/2$ and $r_U = r - \delta/2$. Then

$$N^* = r(F - U) + \frac{\delta}{2}(F + U) + Z^*. \quad (24)$$

A small differential correction is multiplied by the gross sum, not the net difference. Near balance, unequal source and sink corrections can dominate the common-scale contribution and reverse the sign of net exchange. Pairing therefore creates a testable mechanism for hidden gross bias, not an automatic means of making any proposed stock compatible with atmospheric constraints.

The contemporary inventory provides a scale illustration, distinct from a calibration estimate. EDGAR describes positive and negative LULUCF terms each near $6 \text{ GtCO}_2\text{e yr}^{-1}$ and a residual near 0.9 [9]. Relative to their combined magnitude of order 12, the signed-category cancellation ratio is of order $0.9/12 \approx 0.075$ (roughly seven percent). This uses rounded inventory aggregates, not ecosystem gross photosynthesis and respiration, and does not show that either term shares a uniform measurement error.

6.4 Size-dependent calibration and a directional census test

Large-tree structural discrepancies motivate a differential-correction test [3], but they do not by themselves establish that mortality is corrected more than growth. To see the distinction, let x be a tree's baseline carbon and let a stable, continuously differentiable, zero-intercept calibration be $m(x) = r(x)x$. A surviving tree growing from x_0 to $x_1 > x_0$ has gain multiplier

$$q_G(x_0, x_1) = \frac{m(x_1) - m(x_0)}{x_1 - x_0}, \quad m'(x) = r(x) + xr'(x). \quad (25)$$

The secant equals the average of m' over that interval. Mortality of a tree with assigned pre-death stock x_d , in contrast, contributes $m(x_d)/x_d = r(x_d)$. An increasing stock multiplier is not the same function as the growth multiplier. Species, architecture and changing organ allocation can require a multivariate calibration; the scalar construction isolates the logical distinction without imposing a universal fitted curve.

Let $G, L > 0$ denote positive gains and losses in a matched live-carbon census account, with net accumulation $S = G - L$. Sum calibrated survivor increments and consistently accounted recruitment into G^* ; use matched pre-death stocks and other specified live-pool losses for L^* . The resulting ratios $r_G = G^*/G$ and $r_L = L^*/L$ give

$$S^* = r_G G - r_L L = r_G S - (r_L - r_G)L, \quad S^* - S = (r_G - 1)G - (r_L - 1)L. \quad (26)$$

Recruitment across an inventory threshold is an entry into the observed pool, not necessarily carbon assimilated wholly during that interval. Diameter thresholds, survivor shrinkage, missing stems and death-time conventions must therefore be fixed or reconciled. Mortality here is loss from the live pool; atmospheric release follows its carbon-fate operator, not immediate emission by definition.

Directional hypothesis H_Δ . In a specified, independently calibrated census population, the event-weighted loss correction exceeds the gain correction by a preregistered meaningful margin, $r_L - r_G > \epsilon_\Delta \geq 0$. This predicts $S^* < r_G S$, lower net live-carbon accumulation than under common gain scaling. For S^* to be smaller than the *uncorrected* sink requires the stronger inequality $(r_L - 1)L > (r_G - 1)G$. Neither prediction follows solely from larger average size among dead trees. The hypothesis concerns measured event-weighted corrections. A prespecified interval for their difference wholly above the margin supports the directional claim; an interval wholly at or below the margin opposes it, while overlap leaves the test unresolved at that precision.

Table 5: Synthetic differential-census cases with $G = 120$, $L = 100$ and $S = 20$ in arbitrary carbon units over a common interval. The second row uses one increasing size-calibration function, not fitted forest observations.

Case	r_G	r_L	G^*	L^*	S^*
Loss-weighted hypothesis	1.20	1.50	144.0	150.0	-6.0
Increasing-size counterexample	1.85	1.50	222.0	150.0	72.0

The counterexample in Table 5 uses $m(x) = x(1 + 0.005x)$ in the specified synthetic units: 12 survivors grow from 80 to 90, while one larger tree of stock 100 dies. Then $r_G = 1.85 > r_L = 1.50$ and $S^* = 72$, despite the strictly increasing $r(x)$ and the larger dead tree. Underestimation of large stock levels therefore does not determine the corrected sink's direction. This counterexample does not deny differential amplification; it identifies the finite-increment evidence needed to turn that mechanism into a defensible prediction.

6.5 Exact size criteria and calibration-family-specific screening

Assume a fixed scalar calibration applies to every event in a matched census. Let j index positive growth intervals $[x_{0j}, x_{1j}]$, $g_j = x_{1j} - x_{0j} > 0$, and d mortality stocks $x_d > 0$. Set $G = \sum_j g_j$, $L = \sum_d x_d$, with weights $v_j = g_j/G$ and $\ell_d = x_d/L$. Recruits may enter with $x_{0j} = 0$ under the specified observed-pool convention. Shrinkage and other losses require additional signed terms, not an unnoticed change in the event set.

Proposition 6.3 (Exact quadratic census criterion). *For $m(x) = \alpha x + kx^2$, with $\alpha > 0$ and $k > 0$ in consistent carbon units, define*

$$\bar{x}_L = \sum_d \ell_d x_d = \frac{\sum_d x_d^2}{L}, \quad \bar{x}_G = \sum_j v_j \frac{x_{0j} + x_{1j}}{2}. \quad (27)$$

Then

$$r_L - r_G = k(\bar{x}_L - 2\bar{x}_G). \quad (28)$$

Thus $r_L > r_G$ exactly when $\bar{x}_L > 2\bar{x}_G$; a positive practical margin requires $k(\bar{x}_L - 2\bar{x}_G) > \epsilon_\Delta$.

Proof. For a death, $m(x_d)/x_d = \alpha + kx_d$. For a gain, the exact secant is $\alpha + k(x_{0j} + x_{1j})$. Weighted summation and subtraction prove the result. Neither a small-increment limit nor zero recruitment is assumed. $\square$

The constant scale α cancels from the difference and positive k from its sign, but not from the effect magnitude. The counterexample in Table 5 is exactly $100 < 2 \times 85$. This criterion can screen that calibration family from baseline event data before fitting k ; it is not a calibration-free test of all biological models.

For the power family $m(x) = ax^b$, $a > 0$, $b > 1$ (with the units of a chosen consistently), a finite-interval extension avoids relying on infinitesimal growth. Put $p = b - 1$, and define a probability distribution γ along the gain intervals by

$$\mathbb{E}_\gamma[h(X)] = \frac{1}{G} \sum_j \int_{x_{0j}}^{x_{1j}} h(x) dx, \quad M_{p,L} = (\mathbb{E}_\ell[X^p])^{1/p}, \quad M_{p,G} = (\mathbb{E}_\gamma[X^p])^{1/p}. \quad (29)$$

Overlapping intervals represent distinct tree events and are counted separately.

Proposition 6.4 (Exact finite-interval power-law criterion). *For these event distributions,*

$$r_L = aM_{p,L}^p, \quad r_G = abM_{p,G}^p, \quad r_L > r_G \iff \frac{M_{p,L}}{M_{p,G}} > b^{1/(b-1)}. \quad (30)$$

Proof. The loss formula follows by weighting ax_d^{b-1} . The fundamental theorem of calculus gives $x_1^b - x_0^b = b \int_{x_0}^{x_1} x^{b-1} dx$, proving the gain formula. Positive division and the $1/p$ power preserve the inequality. $\square$

The threshold decreases from e as $b \downarrow 1$ to 2 at $b = 2$. Indeed, $\log[b^{1/(b-1)}] = \log(1+p)/p$ decreases for $p > 0$. At $b = 1$, exactly, both factors are a and their difference is zero; the limiting threshold is not a strict-direction test at that endpoint. In the small-increment limit, γ reduces to the event-gain-weighted size distribution. Its p -power mean, not an arithmetic mean substituted at every exponent, remains essential. At $b = 2$, $M_{1,G} = \bar{x}_G$ and the threshold is exactly the quadratic one.

A single arithmetic ratio below 2 does *not* exclude all $1 < b \leq 2$ in heterogeneous censuses. For example, let nine survivors grow from 1 to 1.01, one from 500 to 500.01, and one stock-100 tree die. The ratio $\bar{x}_L/\bar{x}_G = 100/50.905 \simeq 1.9644$ fails the quadratic screen. Nevertheless, with $m(x) = x^{3/2}$ the exact factors are $r_L = 10$ and $r_G \simeq 4.70749$, so losses are corrected more. This is a synthetic counterexample, not a plausible-sample assertion. The valid screening statistic is exponent-specific: compute the left side of (30) for the prespecified family. Excluding an entire exponent interval requires the inequality to fail throughout that interval, with numerical tolerances controlled; a few sampled exponents do not prove uniform exclusion. Neither sign screen supplies the calibrated magnitude or independent field support required for H_Δ .

Near-linear calibration. A dimensionally explicit power calibration is $m_b(x) = ax_*(x/x_*)^b$, where $x_* > 0$ is fixed. For $b = 1 + \varepsilon$ and finite logarithmic moments under the loss and gain distributions,

$$r_L - r_G = a\varepsilon [\mathbb{E}_\ell \ln(X/x_*) - \mathbb{E}_\gamma \ln(X/x_*) - 1] + O(\varepsilon^2). \quad (31)$$

This follows by expanding $(X/x_*)^\varepsilon$ and $(1 + \varepsilon)(X/x_*)^\varepsilon$ in the exact loss and interval-gain operators. Recruitment intervals starting at zero are admissible because $\int_0^h |\ln x|^k dx$ is finite for fixed $h > 0$ and finite k . The differential tends to zero linearly with $b - 1$. Its first-order sign is set by the ratio of loss- and gain-weighted geometric means relative to e ; the reference stock cancels in their log difference. If the coefficient vanishes, higher-order terms determine the sign.

6.6 Gross-source and net-land shares are different quantities

For a simplified account with other positive sources O and $Z = 0$, define

$$s_F^{\text{gross}} = \frac{F}{O + F}, \quad s_N^{\text{net}} = \frac{F - U}{O + F - U}. \quad (32)$$

The first is the forest fraction of the selected positive source sum. The second is a signed net-land contribution relative to a net total; it can be negative and becomes unstable as that total approaches zero. $F/(O + F - U)$ is a third ratio: a gross component divided by a net total. It is not a conventional composition of non-negative shares and can exceed one. When Z contains additional sources and removals, those components must be explicitly partitioned before forming a gross-source sum.

Table 6: Synthetic paired-correction example: $F = 6$, $U = 5.8$, $O = 50$, $Z = 0$. Flow units are arbitrary common annual mass units, not 2025 observations. The last two columns follow (32); percentages have different denominators.

Case	r_F	r_U	F^*	U^*	$F^* - U^*$	Gross share (%)	Net share (%)
Reference	1.0	1.0	6.00	5.80	0.20	10.71	0.40
Source only	2.5	1.0	15.00	5.80	9.20	23.08	15.54
Paired uniform	2.5	2.5	15.00	14.50	0.50	23.08	0.99
Paired differential	2.5	2.6	15.00	15.08	-0.08	23.08	-0.16

In the uniform example, corrections of +9 and +8.7 in gross emission and removal magnitudes change the net by only +0.3. The cancellation ratio is about 0.01695. With a slightly larger sink multiplier, the

revised net becomes negative. These are algebraic demonstrations of identifiability and denominator dependence, not inferred global forest fluxes. They motivate independent constraints on source and removal weights, rather than a preferred multiplier.

6.7 Independent measurements remain decisive

A common allometric calibration can affect census-derived quantities, but it need not affect an independently measured atmospheric concentration increment, fossil-fuel statistic or tracer constraint in the same way. The theoretical cancellation concerns shared calibration of specified components; regrowth and live-census accumulation enter only after their increment operators and pool boundaries have been aligned. It cannot be imposed on independently constrained terms merely to force closure.

Likewise, living-biomass uptake, managed-land net removals and bookkeeping net land-use emissions are not interchangeable sink variables. A correction must follow the underlying mass estimator into its valid pool and process, account for time lags, and identify overlap with other entries. Only then can the proposed source–sink revision be tested against a carbon-mass budget. The fact that a budget can contain compensating errors is not evidence that any particular compensation occurs [10].

7 Accounting alternatives and the historical percentage audit

7.1 Three alternatives retained

For a harmonized signed account $T = F + O - U$, with other signed terms absorbed into the defined O , the governing adjustment is

$$\Delta T = \Delta F + \Delta O - \Delta U. \quad (33)$$

Alternative I: retain other contributions. If $\Delta O = \Delta U = 0$, a higher F raises T . Unchanged sectors have lower shares of the enlarged total but unchanged absolute emissions; their ranking relative to each other is preserved.

Alternative II: retain the total and removals. If $\Delta T = \Delta U = 0$, then $\Delta O = -\Delta F$. That is an aggregate condition, not evidence about which individual sectors should change. Uniform proportional reduction is an additional allocation rule, not a consequence of the balance.

Alternative III: jointly reassess components. Sources, removals and total may change together. The paired correction in Proposition 6.1 is a structured instance, not unrestricted uncertainty. Independent evidence can accept, constrain or reject it. None of the alternatives is selected solely by the shape of a ribbon.

7.2 Tracing the 10.3% benchmark

The historical preprint's emissions discussion cites Ecofys's *World GHG Emissions Flow Chart 2010*, issued in 2013 [1, 8]. The original chart states a global total of 48,629 MtCO₂e, with all percentages referring to that total. The 10.3% label belongs to *deforestation / afforestation and land-use CO₂*. It is therefore a combined land-use category, not a separately identified gross tropical-deforestation flow.

Its displayed share implies approximately $0.103 \times 48.629 \approx 5.01$ GtCO₂ for that CO₂-labelled component. This is a calculation from rounded graphical inputs, not precision established by independent data. Because the numerator is CO₂, using the multi-gas denominator is dimensionally permissible; the substantive problem is category and source compatibility, not the mere presence of CO₂ inside a CO₂e chart.

The chart cites IEA, EDGAR, CDIAC and Global Carbon Project source vintages. It does not specify a recoverable link from the 10.3% numerator to either the 228.7 or 285.1 PgC stock map, and its plate does not identify a unique GWP assessment basis. The cited plate is the original Ecofys-authored chart

preserved by an external mirror. Its original publisher URL and retrieval limitations are recorded in the provenance notes; no unverified Wayback capture is substituted for an accessible source. The exact component assembly and spatial density correspondence therefore remain unverified. These omissions are not filled by assuming that all contemporaneous land-use estimates used the same carbon-density model.

Consequently, multiplying 10.3% by the historical stock ratio cannot presently be interpreted as a corrected gross-deforestation percentage. The baseline combines removals and land-use effects; the stock domains differ; and the numerator's density provenance is not matched. Each problem is logically prior to choosing a fixed or variable denominator. The displayed historical arithmetic is retained to diagnose the original inference, not to preserve it as a newly supported empirical result.

7.3 Two arithmetic reconstructions, not two measurements

Let $s_0 = 0.103$ denote the chart's combined category share, and provisionally assign it a scalar multiplier r_L . For the arithmetic demonstration only, set $r_L = r_C$ using the displayed stock ratios:

$$r_{C,1} = 723.97/285.1, \quad r_{C,2} = 723.97/228.7. \quad (34)$$

The fixed-total calculation and the retained-other-contributions calculation are respectively

$$s_{\text{fixed}} = r_L s_0, \quad s_{\text{renorm}} = \frac{r_L s_0}{1 + (r_L - 1)s_0}. \quad (35)$$

The latter follows by adding the component increment once to the total. The former requires an offset elsewhere to retain that total. Neither establishes that the combined historical land-use component has the imposed multiplier.

Table 7: Reconstruction from displayed historical inputs. The original published range is 26.15–32.60%; nearest rounding of the displayed inputs gives 26.16–32.61%, whereas truncation gives the published 26.15–32.60%. No endpoint is a confidence limit. $r_L = r_C$ is imposed despite unverified spatial and numerator compatibility.

Comparison stock (PgC)	Ratio	Fixed total (%)	Retained others (%)
285.1	2.53935	26.16	22.58
228.7	3.16559	32.61	26.66

The renormalized interval is 22.58–26.66%. The historical fixed-total endpoints are compatible with at least two display paths: truncating the calculation with a baseline of exactly 10.3%, or rounding a calculation with an unprinted baseline share of approximately 10.296665–10.299824%, which itself displays as 10.3%. The percentage-column pattern favors truncation as a coherent explanation, but the source calculation has not been recovered to identify its actual path. Both possibilities are recorded rather than assigning the endpoint difference to an established cause. This distinguishes denominator arithmetic from the much harder stock-to-flux inference. For a harmonized net land component, uniform paired scaling could produce a common multiplier of its calibrated net part, as (23) shows. That mathematical possibility does not establish that Ecofys's component, the historical map totals and the relevant removal estimates meet those conditions.

Figure 2 therefore puts the two ranges in a labelled historical inset outside the forest ribbon. The green branch identifies forest-carbon remeasurement and paired source–sink revision. The surrounding question marks denote incomplete allocation, not license to transfer an arbitrary amount from energy or any other named sector. The 2010 inset is not a numerical branch of the 2025 inventory.

8 Identification and an empirical testing programme

8.1 What a carbon budget can identify

An atmospheric carbon constraint takes a carbon-mass form such as

$$G_{\text{atm}} = E_{\text{fossil}} + F - U - S_{\text{ocean}} + R, \quad (36)$$

where terms must use the same domain, period and sign convention, and R contains specified additional net processes or residuals. It is not valid to substitute a multi-gas GWP-weighted total for the carbon source sum. With all other terms fixed, this one balance constrains $F - U$, not F and U separately. The transformation $(F, U) \mapsto (F + a, U + a)$ leaves it unchanged wherever the revised components remain physically admissible.

This non-identifiability does not imply equal support for every admissible pair. Regional and temporal observations, repeated biomass measurements, isotopic or co-emitted tracers, fire observations, ocean constraints and carbon-fate data can narrow the null direction. Atmospheric profiles can provide information different from bottom-up allometry [27]. The relevant question is how much independent information survives shared calibration and transport-model uncertainty, not whether one balance can close.

The fixed-operator analysis also implies a design priority: estimate the joint distribution of local corrections and both source and sink weights. A larger number of intact plots alone need not identify the release-weighted correction in forests being disturbed. Likewise, a clearing-focused sample cannot by itself identify the correction to regrowth or intact-forest accumulation.

8.2 Four linked empirical tests

Measurement and density. Representative strata should cross forest type, disturbance status, tree-size distribution and soil–climate setting. Independently validated reconstruction and destructive-reference opportunities should test material volume, basic density, carbon fractions and organ completeness. Validation must include the high-biomass tails required by the historical hypothesis, rather than excluding inconvenient large trees or treating model emulation as validation. No new destructive sampling of intact old-growth forest is prescribed solely for this purpose; existing harvest records, legally available material and independently cross-checked non-destructive measurements can supply reference information.

Common spatial support. Reintegrate the original and comparison surfaces over an identical mask, resolution, year and carbon-pool definition. Preserve fractional forest cover and degraded vegetation explicitly. Audit units and row metadata before computing ratios. A model that explains differences chiefly by mask or moisture conventions should not have those differences relabelled as an allometric correction.

Paired annual transfer. Estimate disturbance cohorts, release timing, retained products, dead-pool transfers and regrowth. Calibrate repeated-census gains and losses with the same measurement model where justified, while allowing different correction distributions for source and removal strata. Report r_C , operator-derived r_F and r_U , applicable covariance terms and the cancellation ratio, not only a global mean stock multiplier. Resolve tree-size and organ compartments in initial burning, later decomposition and products, and test the ordering and cumulative-boundary conditions in Section 5.4. Test whether temporal calibration drift in (11) can explain an apparent flux change.

Independent reconciliation. Compare the proposed gross and net changes with a harmonized carbon budget and independent regional observations. Specify tolerances, comparison margins and model-rejection rules before selecting a favoured allocation. Where only a net constraint is identifiable, report a constrained family of source–sink pairs rather than a falsely unique decomposition. A paired correction that requires unsupported compensation in independently constrained terms should fail that test.

For H_Δ , first compute the family-specific event-size diagnostics in Section 6.5; record whether the proposed directional sign is even admissible under that family. Then estimate the calibration on reference

trees separate from the census-event evaluation, with site or landscape holdouts to avoid treating correlated stems as independent validation. Apply the fixed calibration to both endpoints of surviving trees and to the assigned stocks of mortality events; include recruits with a stated threshold convention. Estimate r_G , r_L and their joint uncertainty from those event-weighted sums, not from average size-class correction factors. Prespecify the spatial domain, census interval, margin ϵ_Δ and uncertainty criterion; propagate shared calibration and sampling dependence, and report both $S^* - r_G S$ and $S^* - S$. A larger loss correction in that population would support the directional prediction. A larger marginal growth correction, as permitted by (25), would falsify that direction without falsifying the general transfer identities.

8.3 Discriminating outcomes

Three outcomes would materially strengthen the measurement argument: a positive independently measured carbon correction in representative high-emission strata; a validated covariance structure linking that correction to disturbance and uptake; and agreement of the resulting net and regional changes with independent observations. Equally informative outcomes could show correction concentrated in retained rather than disturbed forest, downward chemical adjustments offsetting geometric changes, or differences explained largely by spatial support.

A large common gross correction with nearly cancelled net effects is one testable outcome, not a prerequisite. If source and sink corrections differ enough to produce the amplification in (24), the resulting net signal must be observable within the relevant uncertainty. Neither the absence of a large global residual nor the existence of regional allometric underestimation alone resolves those alternatives.

The Global Carbon Budget identifies low confidence in its 2025 land-use projection and limitations in capturing degradation from selective logging, edges and fragmentation within that projection [10]. These limitations motivate process-specific observations, not an arbitrary global increment. Independent assessments of Amazonian degradation likewise distinguish mechanisms and affected areas rather than replacing them with one factor [28].

9 Discussion and conclusions

The measurement challenge is stronger when separated from a commitment to one unusually high historical stock. The 723.97 PgC extrapolation has a demanding density implication that is not explained away by its larger mixed vegetation domain: its high-carbon classes alone exceed the comparison stock on less area. The crossed class intervals suggest a specific metadata test but do not establish a corrected raster. Its comparison with earlier totals does not identify pure structural bias. The historical 10.3% benchmark additionally refers to a combined land-use CO₂ category within a multi-gas total, without a verified correspondence to the compared stock surfaces. Those findings prevent an empirical reinterpretation of either historical percentage range as a newly established gross forest-loss share.

None of these findings makes conventional biomass estimates immune to revision. Harvest comparisons and local calibration studies support testing structural bias, while counter-direction results and carbon chemistry require its sign and magnitude to remain empirical. The transfer identity formalizes the critical association between correction and annual release. The paired extension adds a second association with removals: shared calibration can conceal gross changes in a net balance, but heterogeneous corrections can destroy that cancellation. Regrowth requires gain-weighted rather than initial-stock-weighted transfer. For live-census sinks, size-dependent stock corrections act through marginal growth and mortality levels differently, so a directional claim must be tested at the event level. Quadratic and finite-interval power-law criteria identify family-specific signs before fitting the bias magnitude; an arithmetic size ratio is not a universal exclusion test. Oppositely ordered release weights can also delay the expression of a structural correction, with cumulative recovery only under a matched ultimate-fate boundary.

The principal operational conclusion is therefore not that a particular global total should be enlarged, nor that another sector's emissions should automatically shrink. It is that *biological recalibration must be propagated through matched source and removal processes before its effect on attribution can be determined*. The three original accounting alternatives remain available, now constrained by explicit measurements and boundaries.

Communication should expose that structure. A small net land residual is not a direct measure of a small forest-loss source. A large standing stock is not an annual emission. A gross component divided by a net total is not a conventional source composition. An enlarged green ribbon can identify an important question, but it cannot resolve these distinctions by visual prominence. The diagrams separate measured reference quantities from hatched, unresolved corrections and identify conditional numbers explicitly.

Carbon storage, annual exchange and hydrological functions also remain distinct ecological quantities. Evidence that deforestation affects tropical precipitation provides an independent mechanism of ecosystem consequence [29]; it does not depend on establishing tropical deforestation as the largest annual greenhouse-gas source. Accurate conservation science does not require those different mechanisms to be reduced to one percentage.

The resulting framework is independent of the historical numerical claim. Its stock-to-flux and paired source–sink identities are exact under stated assumptions; its audit calculations are reproducible; and its empirical programme can support, localize or reject a proposed correction. What remains unresolved is the independently validated, spatially matched distribution of forest-carbon bias and its annual transfer—not the algebra governing that transfer.

Data and code availability

The manuscript, Overleaf source, transcribed inputs, provenance records, editable figures and numerical outputs are available as [research release 5.2](https://github.com/hyrucañji77/Tropical_Forest_Carbon-) in https://github.com/hyrucañji77/Tropical_Forest_Carbon- [11]. Release metadata, the Git commit and the SHA-256 manifest identify the snapshot. `data/source_provenance.csv` documents sources and inference limits; `data/display_precision_audit.json` gives both display-error models; `verification.json` records package checks. Running `python scripts/reproduce.py` regenerates the numerical tables without network access. The deposit supports published-data and analytical reproduction; unrecovered original records are described in `SOURCE_AVAILABILITY.md`. No archival DOI has been assigned.

Author contribution

Henry Arellano-Peña is the sole author of this manuscript and is responsible for its analysis, interpretation and presentation.

Competing interests

Henry Arellano-Peña owns the methodologies examined and is founder and director of NEBIOT S.A.S., the sole company currently affiliated with this work. NEB001 supplies the CCA–Fuzzy Land Cover vegetation mapping and NEB002 the voxel, elemental-carbon and neural-network quantification chain underlying the historical estimate audited here [1, 30]. These ownership and professional relationships are disclosed; the author declares no other competing interests.

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Use of generative artificial intelligence

ChatGPT (OpenAI) assisted with drafting, mathematical exploration, source discovery, code development and LaTeX preparation. Claude (Anthropic) generated critical-review material that informed revisions. Suggestions from both systems were assessed against the stated mathematical derivations and identified sources rather than treated as empirical evidence or bibliographic authority. Computational checks test specified identities and calculations and do not replace independent biological measurements. Responsibility for the scientific content and submitted declarations remains with the named author.

A Compact statistical and methodological notes

A.1 Reconciliation and structural identifiability

For an observation model $y = Hx + \epsilon$ with parameter covariance or likelihood specified independently, a linear combination $a^T x$ is identifiable from noiseless Hx exactly when a belongs to the row space of H . To see this, two parameter vectors with the same observation differ by $v \in \ker H$. Their target values agree precisely when $a^T v = 0$ for every such v , equivalently $a \in (\ker H)^\perp$. This is the general form of the source–sink ambiguity in (36).

Under an independently specified Gaussian reference x_0 with covariance Σ and an observation-error covariance V , the unconstrained conditional mean is

$$\hat{x} = x_0 + \Sigma H^T (H \Sigma H^T + V)^{-1} (y - H x_0). \quad (37)$$

This standard result does not add information absent from the covariance and observations. Hard physical bounds or non-Gaussian discrepancies require an appropriate constrained or non-linear analysis. A statistically unique posterior mean produced by a prior is not the same as structural identification by the data. Shared allometric calibration belongs in covariance or explicit model-discrepancy terms, not in a repeated claim of independent evidence.

A.2 Bias, covariance and ratios

For a squared-error estimator, $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$. Additional samples can reduce sampling variance without eliminating a common model error. Inventory uncertainty may involve activity data, emission factors and correlated aggregation [31]. The paired correction is a structured instance of such dependence, not a claim that all uncertainties are equal or symmetric.

For a ratio $s = F/T$ with nonzero T , first-order propagation gives

$$\text{Var}(s) \simeq \frac{\text{Var}(F)}{T^2} + \frac{F^2 \text{Var}(T)}{T^4} - \frac{2F \text{Cov}(F, T)}{T^3}. \quad (38)$$

If T contains F , independence is generally unjustified. The approximation is unreliable near a zero or poorly constrained denominator. Joint simulation or explicit ratio inference is then preferable. The present article does not fit an uncertainty interval around the historical reconstructed percentages.

A.3 Model domains and admissibility

A unit with zero baseline carbon and zero corrected contribution may be omitted from the restricted fixed-stock release representation. That permission does not extend to regrowth from zero stock, positive corrected flux on a zero baseline increment, or newly represented carbon pools. Those cases require the increment or additive terms in Section 6.2. Newly included areas are a support change and must be accounted for separately. Positive baseline flux is necessary to define a multiplier. Signed net weights do not satisfy the convex-combination bounds for r_F ; separating positive source and removal components avoids that misuse. The general net correction in (17) does not require common calibration; only its

common- r_C expansion does. Separate bases admit two transfer identities, but no common covariance term follows without an explicit mapping. The algebraic identity for a net correction remains valid when its source and removal ratios are supplied by different admissible operators; the common-stock covariance formula does not.

For repeated censuses, additions and removals of sampling units must be reconciled before invoking scalar change identities. For delayed release, the cohort index can absorb spatial position, disturbance year, tree-size compartment, organ and carbon pool. A change in decomposition kernel, combustion completeness or regrowth trajectory is a process correction, not merely a carbon-stock multiplier. These distinctions are necessary both for causal interpretation and for conservation of carbon in an implementation.

B Reproducibility and source audit

B.1 Inputs, outputs and numerical checks

The public repository [11] contains the exact displayed source entries in `data/inputs.json`, including class order, full-precision areas, means, percentages and intervals. It also records the Ecofys category and denominator, the Gonzalez de Tanago validation metric and explicitly synthetic examples. The published table is left unchanged by the diagnostic calculations.

`scripts/reproduce.py`, `scripts/extensions.py` and `scripts/audit_extensions.py` regenerate the numerical and LaTeX-table outputs listed in the manifest. These cover density and support comparisons, crossed intervals, nearest-rounding and truncation models, exact conditional gap probabilities, paired accounts, zero-stock regrowth, finite-increment census criteria, single- and mixed-cohort release, near-linear calibration and invalid inputs. Machine-readable values and typeset tables share the same calculations.

The run uses 1000 baseline cases with seed 20260916, 20,000 extension cases with seed 20260917, and 1100 audit-extension cases with seed 20260918. The total is 257167 assertions, including deterministic regression and invalid-input checks. Algebraic comparisons use absolute tolerance 10^{-10} and relative tolerance 10^{-12} where applicable; decimal interval and conditional-probability calculations use high-precision decimal or exact rational arithmetic. The near-linear expansion is tested through its scaled second-order remainder, rather than asserted exact at finite exponent. Validation records state the tolerance and residual for each family.

B.2 Archive and repository verification

The repository manifest lists relative paths, byte counts and SHA-256 digests for the research snapshot. The verifier checks integrity and compares regenerated numerical outputs within stated cross-platform floating-point tolerances; non-numerical values and CSV/LaTeX text remain exact comparisons. Build reports identify compiler versions, warnings and PDF preflight results. These are integrity and computational-reproduction checks, rather than digital-authenticity certification or biological validation. Source-observation years, publication years and the research-snapshot date are recorded separately.

B.3 Empirical scope and unrecovered records

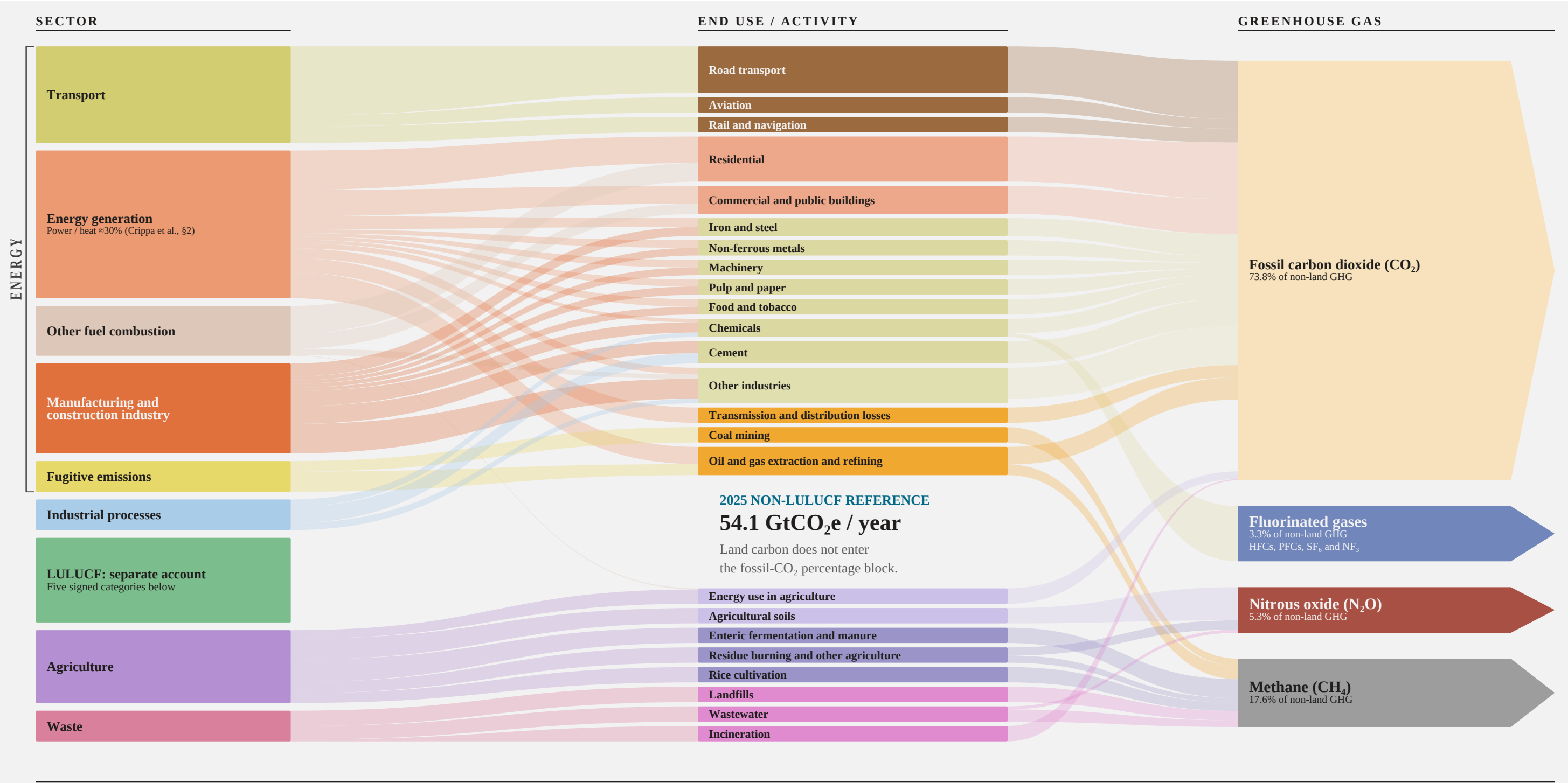
The numerical archive reconstructs published tables and tests exact identities. A common-mask carbon integration, original classification-pipeline rerun, reconstructed Ecofys component inventory and matched field census analysis require additional empirical records. The source-availability note identifies the workbook, raster, class-key and carbon-fraction weights that were not recovered, without assigning an unverified reason for their absence. The display-rule and interval-crossing findings remain diagnostics until those records resolve their computational causes.

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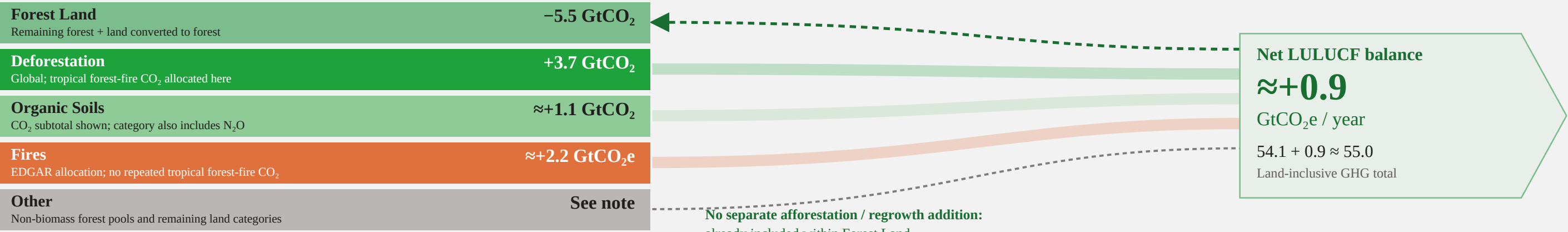
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THE SIGNED LAND ACCOUNT

EDGAR 2026 · Section 3, Figure 6 and Annex 2 · 2025 estimates



Other / rounding remainder: ≈-0.6 GtCO₂e is a closure diagnostic, not an independently extracted category total.

Disjoint land categories, separate gas denominator. Forest Land already includes land converted to forest; no extra afforestation or regrowth subtotal is added. Deforestation includes tropical forest-fire CO₂, which is excluded from Fires; tropical fire non-CO₂ gases remain in Fires. Organic Soils’ quoted 1.1 is CO₂ only. The diagnostic remainder $0.9 - [3.7 - 5.5 + 2.2 + 1.1] \simeq -0.6$ also absorbs omitted subcomponents and rounding; it is not a separately extracted Other estimate. Source: Crippa et al. (2026), booklet Sections 2–3 and Annex 2, doi:10.2760/7717504 (CC BY 4.0).

Figure 1: The reported 2025 account and its boundaries. The sector–activity–gas organization is adapted from WRI (2005) and Ecofys (2013), with author-created vector artwork and a separate five-category signed land account. Intermediate activity nodes describe pathways, rather than EDGAR’s official sector list. Gas shares refer only to EDGAR’s 54.1 GtCO₂e non-LULUCF inventory using AR5 GWP₁₀₀; 73.8% is fossil CO₂. Land emissions therefore do not enter that gas block. The reverse dashed link denotes Forest Land’s net removal. Forest/deforestation estimates use a global Tier 1 approach with default parameters and estimated activity data beyond the latest input maps, not independently remeasured 2025 carbon densities. Paths and all graphical areas are schematic, not a balanced quantitative Sankey. Electricity-attribution paths must not be counted again as direct emissions.

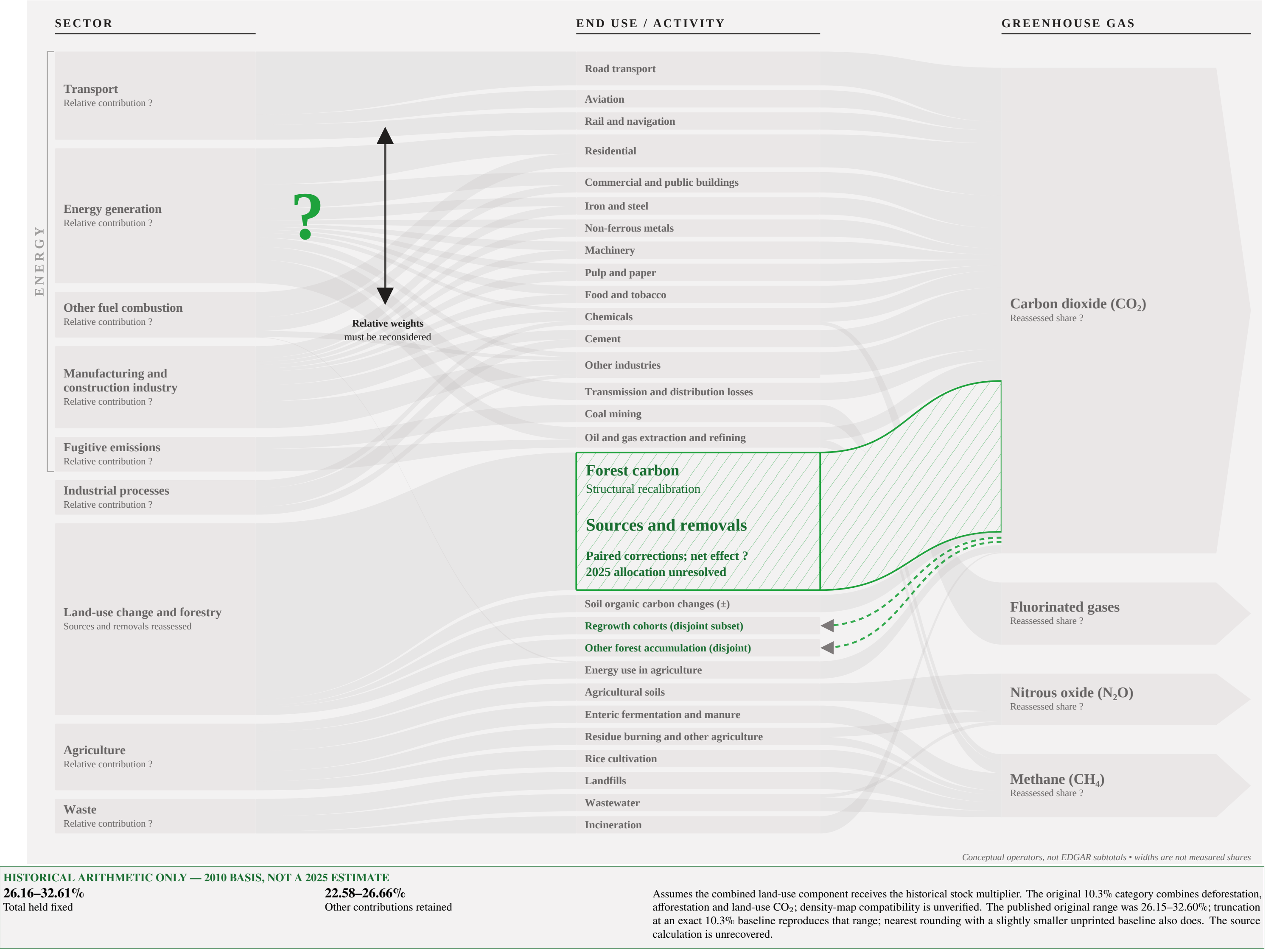


Figure 2: Structural forest-carbon bias and unresolved attribution. The hatched green branch denotes an unresolved remeasurement; its size has no quantitative interpretation. The organization adapts the WRI/Ecofys source–activity–gas convention; reverse removal pathways emphasize the paired analysis; regrowth and other forest accumulation denote disjoint conceptual subsets, not additional EDGAR subtotals. Muted sectors, the question mark and bidirectional arrow retain three alternatives: a larger total, redistribution within a retained total, or joint revision of sources and removals. The inset diagnoses the historical reassessment using the Ecofys (2013) chart; neither interval is an uncertainty bound or a validated correction. The 723.97 PgC extrapolation is audited in Section 3, stock-to-flux transfer in Section 5, paired, regrowth and size-dependent corrections in Section 6, and original-denominator provenance in Section 7.